

AURORA

A Pomodoro Focus Timer with Smart Analytics

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18:42
FOCUS

ABSTRACT

Aurora is a Pomodoro-based productivity application that helps users manage focus and break sessions while providing detailed analytics on their work habits. Built as a modern single-page application, it features an interactive circular timer, a tagging system for categorising work, and a comprehensive insights dashboard with real-time chart visualisations. All data persists locally in the browser, requiring no backend or account setup.

KEY FEATURES

- Interactive Pomodoro Timer** — Circular SVG progress ring with gradient stroke, glow animation, and configurable focus/break durations
- Session Tagging System** — Create and manage project modules to categorise focus sessions with combined module + custom labels
- Analytics Dashboard** — Real-time bar and pie charts showing focus distribution across tags, with Today / Week / All Time filtering
- Persistent Local Storage** — All sessions, settings, and modules persist in localStorage with zero setup and complete data privacy
- Session History** — Reverse-chronological log of every completed focus session with tag, duration, and timestamp

ARCHITECTURE



TECH STACK

React 19 Vite 8 Chart.js ESLint CSS Variables

- React 19.2 with hooks (useState, useEffect, useRef, useLayoutEffect) for state management
- Vite 8 with Oxc plugin for fast HMR and optimised production builds
- Chart.js 4.5 for bar and pie chart visualisations with custom theming

CONCLUSIONS & FUTURE WORK

Aurora successfully delivers a polished, fully client-side Pomodoro experience with rich analytics. The glassmorphism UI with aurora-inspired animations creates an immersive, distraction-free focus environment that encourages sustained concentration.

- Browser notifications and sound alerts on timer completion
- Data export/import for backup and cross-device transfer
- Daily and weekly productivity goals with streak tracking
- Break session analytics and extended session history
- Dark/light theme toggle and additional customisation options
- Cloud sync for cross-device session continuity

DESIGN & UI

Aurora employs a **dark glassmorphism** design with deep purple backgrounds and lavender accents, creating an ambient, distraction-free focus environment. The interface uses backdrop blur effects, gradient overlays, and animated parallax star fields for visual depth.

#120f24 Deep #c5a6ff Accent #e9e2ff Text #fff8ef Heading

KEY METRICS

3

COMPONENTS

367

LINES (APP)

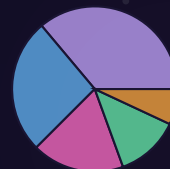
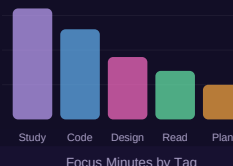
305

LINES (INSIGHT)

0

BACKEND DEPS

DATA VISUALISATION



IMPLEMENTATION DETAILS

- Timer Engine** — useEffect with 1-second intervals and useRef to prevent double session logging. Auto-switches between focus and break modes on completion
- State Persistence** — Initialiser functions read from localStorage on mount; dedicated useEffect hooks sync every state change back. Data survives page refreshes
- Chart Lifecycle** — Chart instances destroyed before recreation on data changes, preventing memory leaks. Custom styling matches the aurora theme palette
- SVG Progress Ring** — Stroke-dashoffset calculations with inline gradients and CSS glow animations for a smooth countdown visual
- Module System** — Case-insensitive duplicate prevention, combined module + tag labelling, and stateful pill-based management UI
- Responsive Layout** — Mobile-first CSS with 768px breakpoint, Inter font stack, and CSS custom properties for consistent theming

SESSION FLOW



Each completed focus session is automatically recorded with its tag, duration, and timestamp. The insights dashboard aggregates this data in real-time, providing bar and pie chart breakdowns of productivity patterns across customisable time ranges.